

# AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

TNC108

# Introduction to serverless computing and AWS Lambda



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

# Serverless computing



# What is serverless computing?

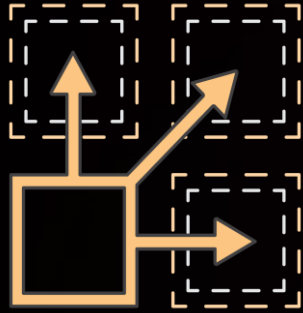
## Traditional deployment and operations

- Provision an instance
- Update OS
- Install app platform
- Build and deploy apps
- Configure auto scaling and load balancing
- Continuously patch, secure, and monitor servers
- Monitor and maintain apps

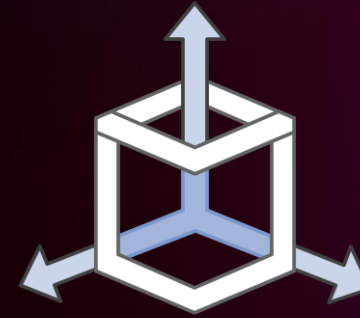
## Serverless deployment and operations

- Provision an instance
- Update OS
- Install app platform
- **Build and deploy apps**
- Configure auto scaling and load balancing
- Continuously patch, secure, and monitor servers
- **Monitor and maintain apps**

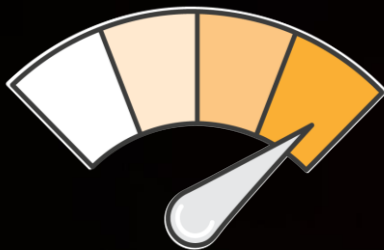
# Benefits of serverless computing



No server management



Continuous scaling

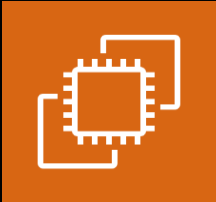


Never pay for idle resources

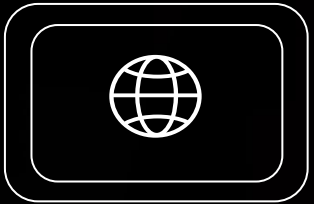


Built-in availability and  
fault tolerance

# Architectural evolution

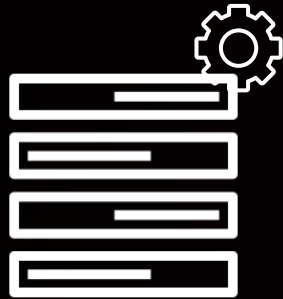
| Category             | Instances                                                                         | Containers                                                                                                                                                              | Serverless                                                                          |
|----------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                      |  |   |  |
| Service              | Amazon EC2                                                                        | Amazon ECS,<br>Amazon EKS                                                                                                                                               | AWS Lambda                                                                          |
| Unit of scale        | Virtual machine                                                                   | Task                                                                                                                                                                    | Function                                                                            |
| Level of abstraction | Hardware                                                                          | Operating system                                                                                                                                                        | Runtime                                                                             |
| Use cases            | Configure servers,<br>storage, networking, and<br>my OS                           | Run microservices,<br>service-oriented<br>architectures in<br>managed containers                                                                                        | Run my code when it<br>is needed                                                    |

# Common serverless application use cases



## Web

- Static websites
- Complex web applications
- Packages for Flask and Express



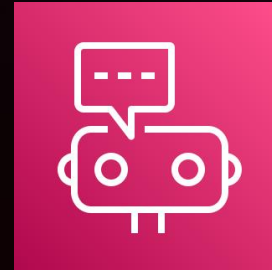
## Backends

- Apps and services
- Mobile
- Internet of Things (IoT)



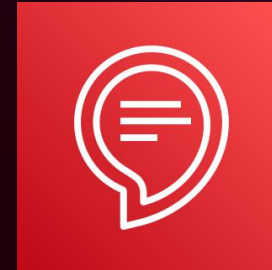
## Data processing

- Real time
- MapReduce
- Batch
- Machine learning inference



## Chatbots

- Chat applications



## Alexa Smart Properties

- Voice-activated apps
- Alexa Skills Kit



## IT automation

- Policy engines
- Extending AWS services
- Infrastructure management

# AWS serverless platform

## Compute



AWS Lambda



AWS Fargate



Amazon API  
Gateway



AWS AppSync

## Storage



Amazon S3

## Database



Amazon  
DynamoDB



Amazon Aurora

## Interprocess



Amazon Simple  
Notification Service  
(Amazon SNS)



Amazon Simple  
Queue Service  
(Amazon SQS)

## Orchestration



AWS Step Functions

## Analytics

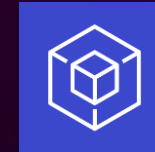


Amazon Kinesis



Amazon Athena

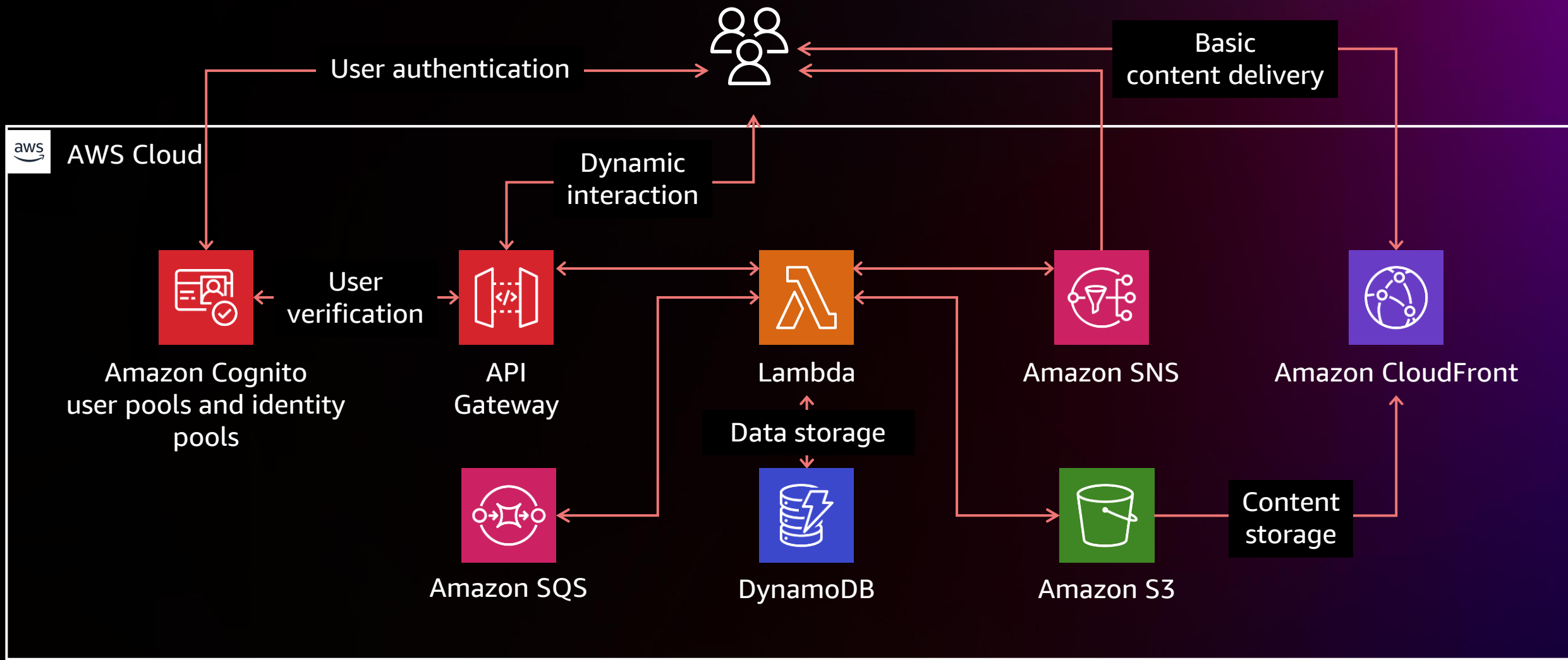
## Developer tools



AWS Tools  
and SDKs



# Typical serverless architecture



# Modern application review



# Analogy: Expanding your food truck to a sit-down diner

What factors influence how you split up the tasks to support the growth of your business?



# Distributed services mean distributed data

## Benefits

- Use **purpose-built** data stores compared to one do-it-all database.
- Optimize for **access and query patterns**.
- **Scale** databases **independently**.

## Challenges

- Accept **eventual consistency**.
- Handle **distributed transactions** and **partial completions**.
- Plan appropriate **extract, transform, and load (ETL)** solutions.

# Event-driven design

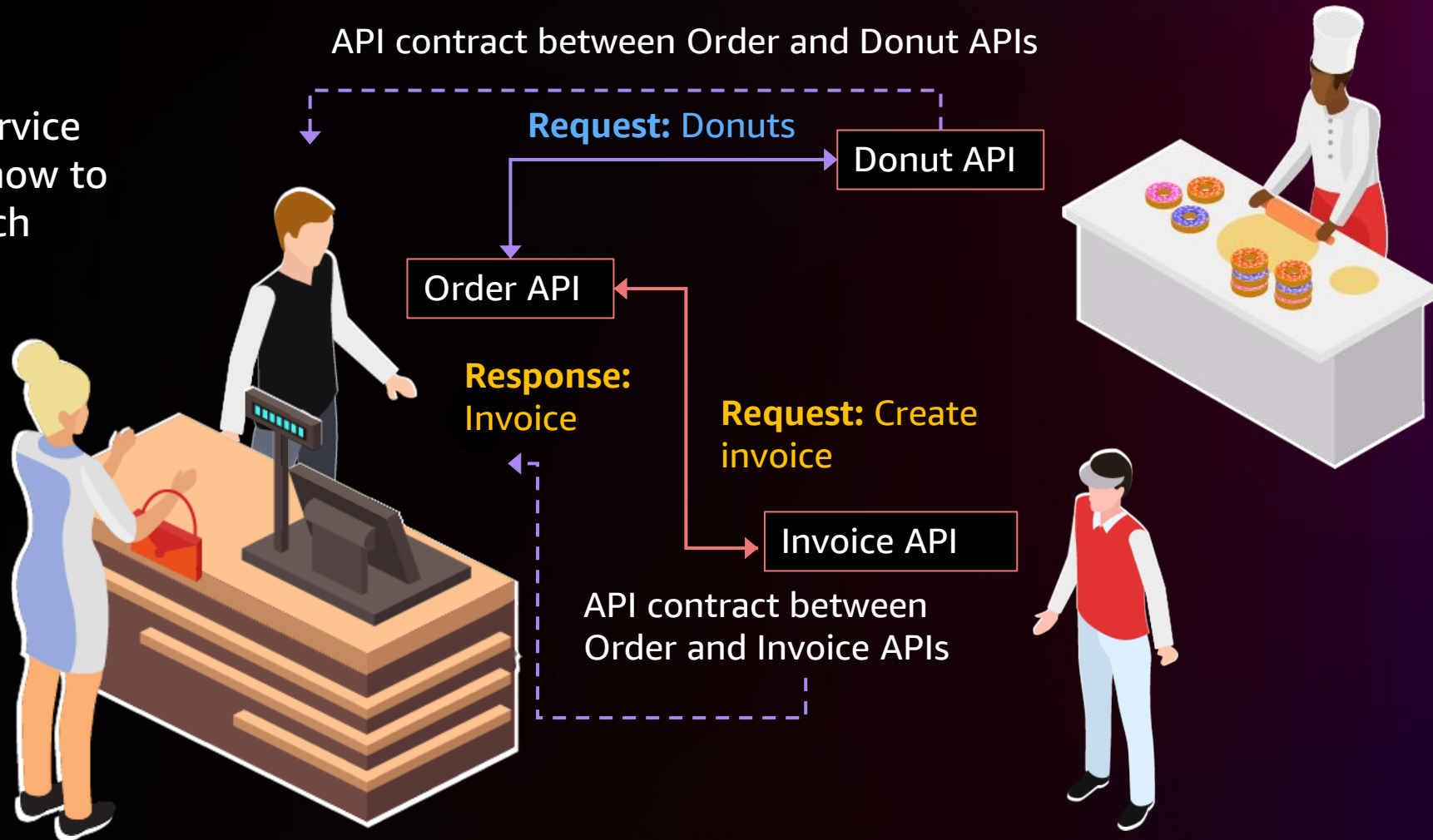


# Communicate through directed commands

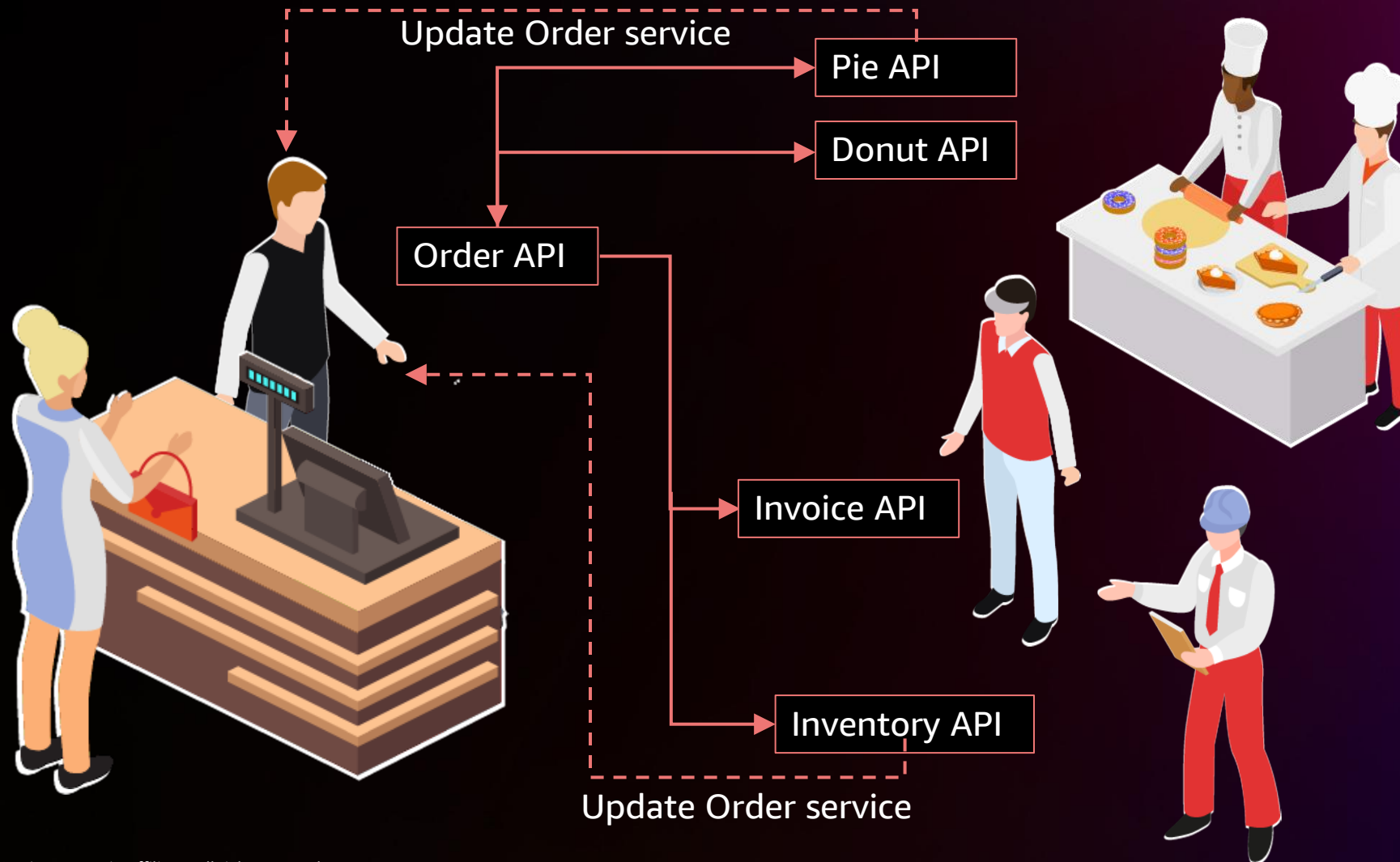


# APIs to communicate directed commands

The producer service needs to know how to interact with each consumer



# Add a consumer and update the producer service

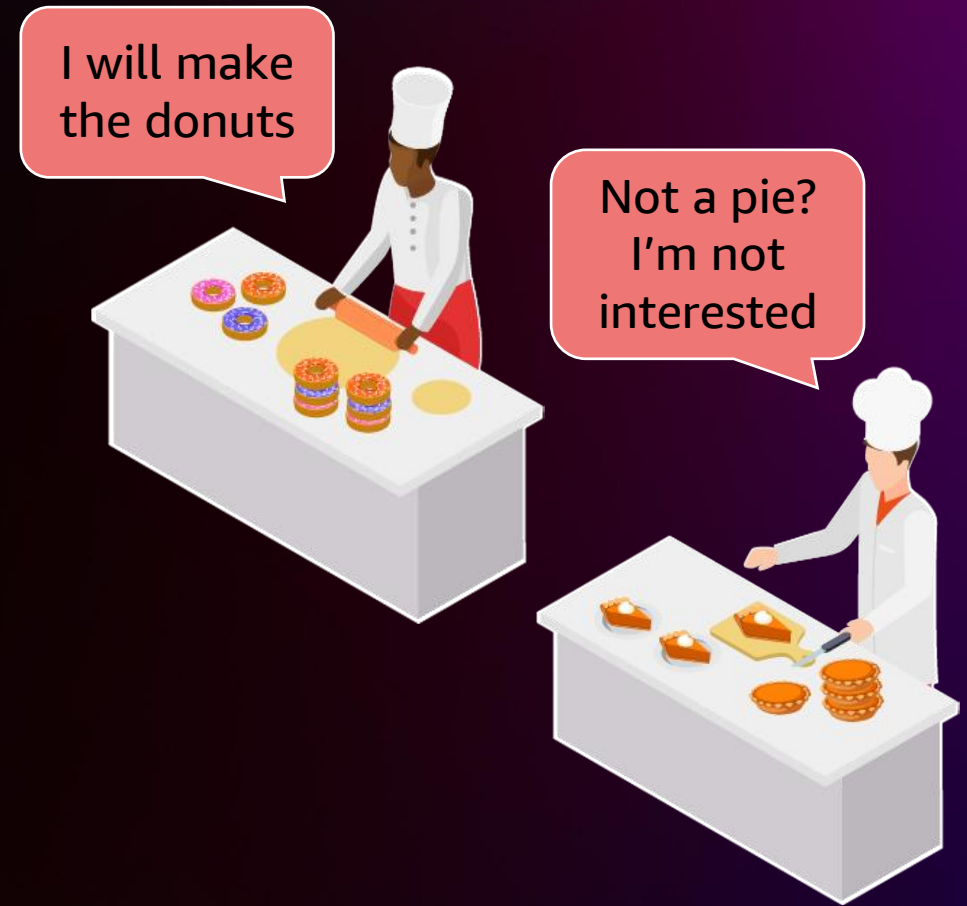




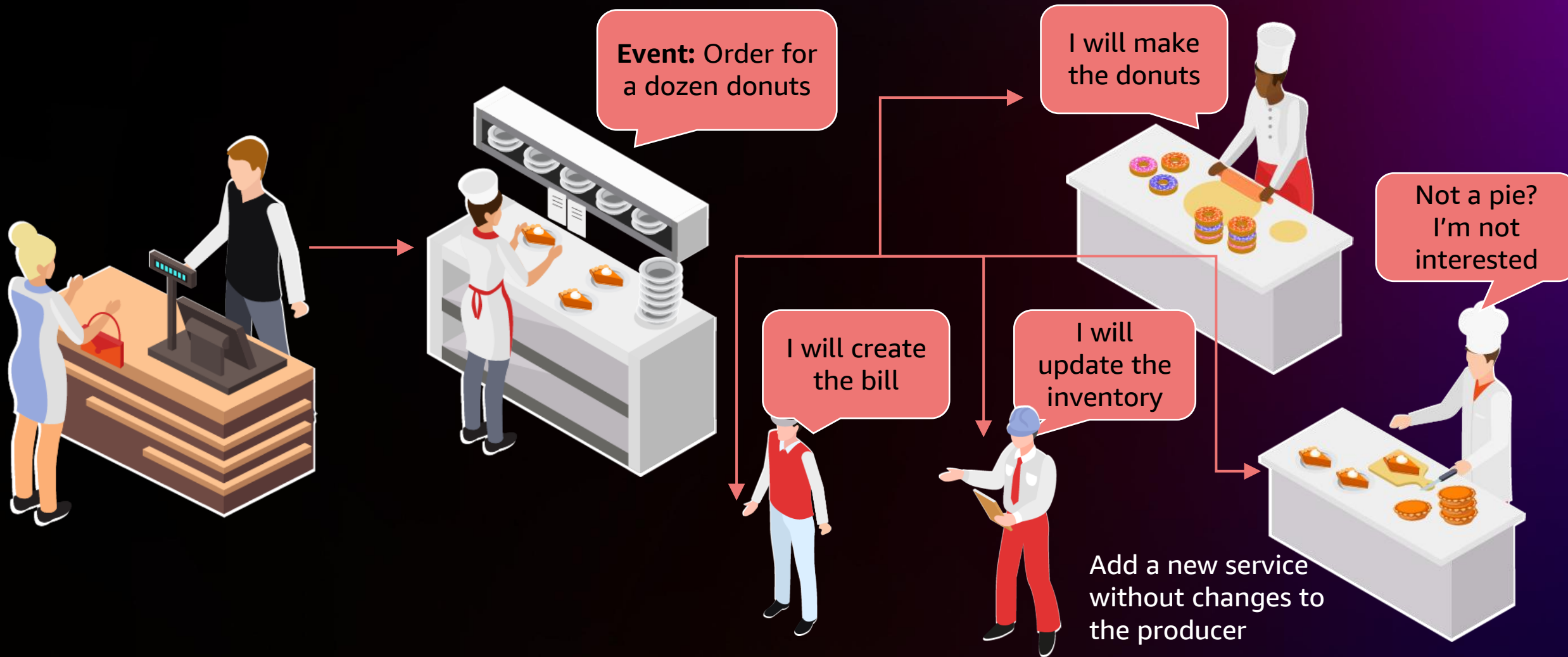
# Events

- Primary mechanism for **sharing information across services** about the **change in state**
- **Observable**, not directed
- Usually processed **asynchronously**

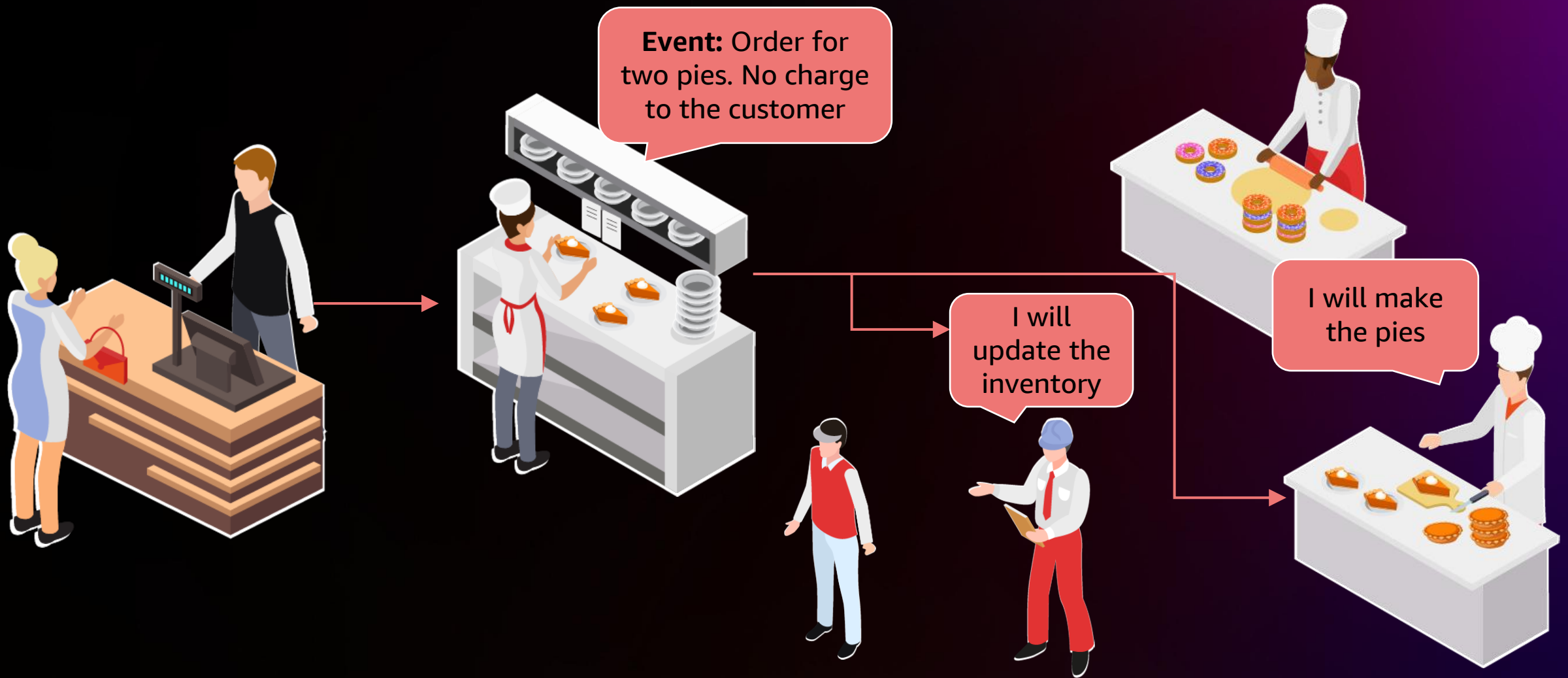
# Communicate by producing observable events



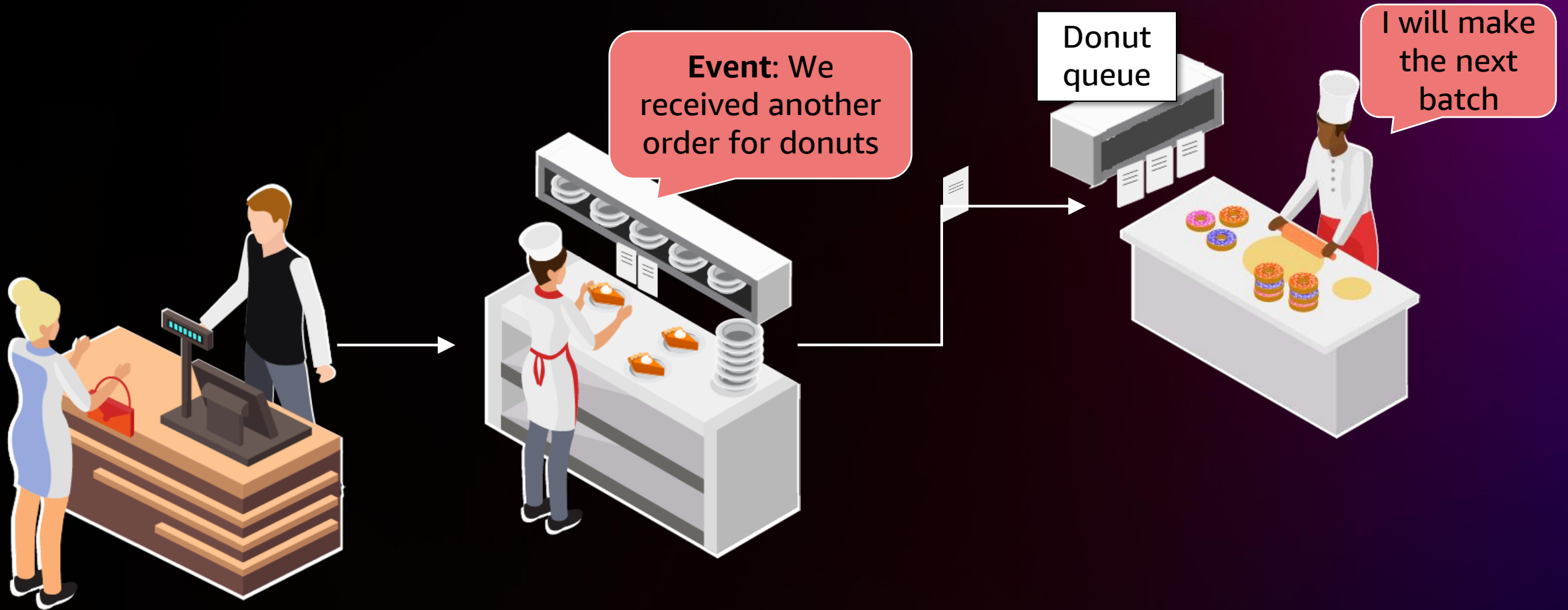
# Share events with multiple consumers



# Routing or filtering of events



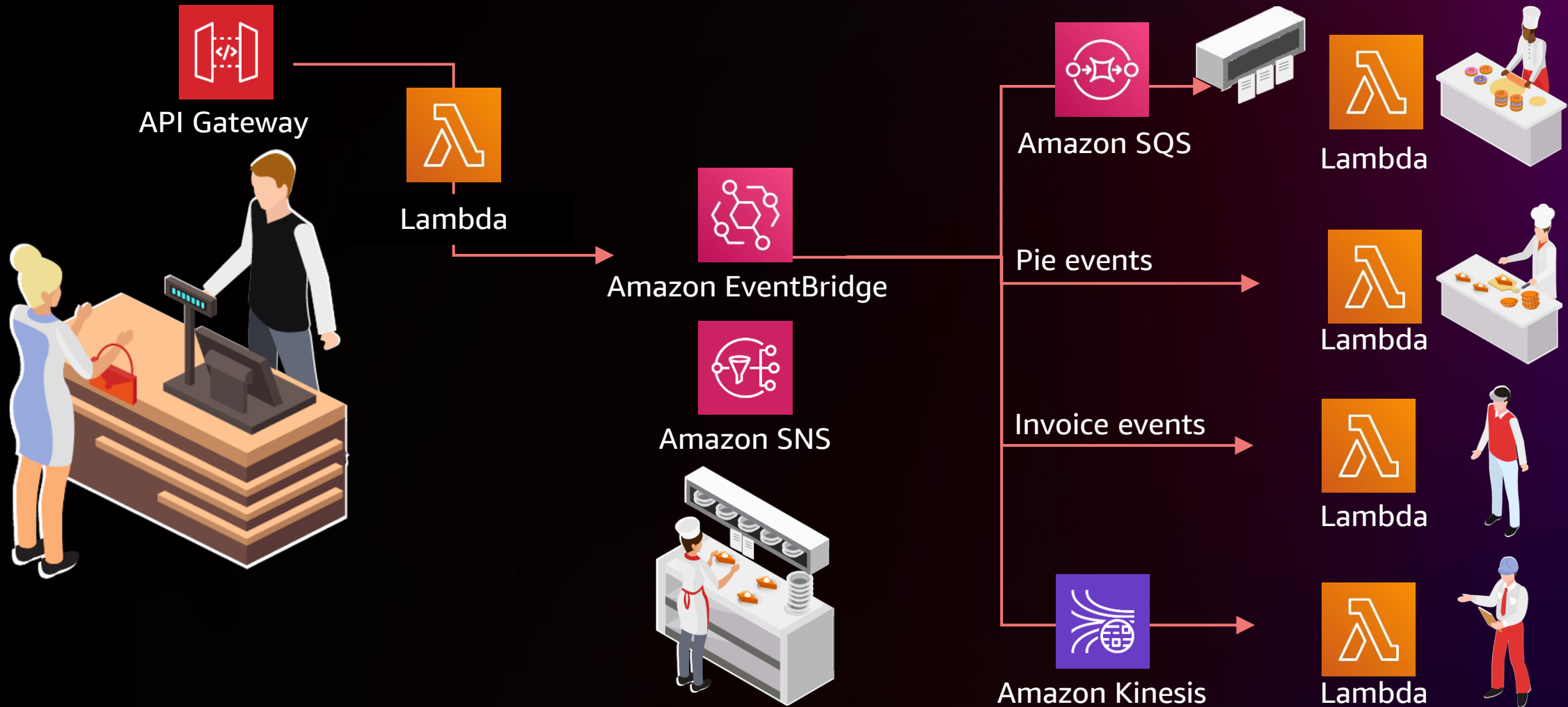
# Provides a buffer to downstream consumers



# Serverless on AWS



# Suitable for modern, event-driven designs



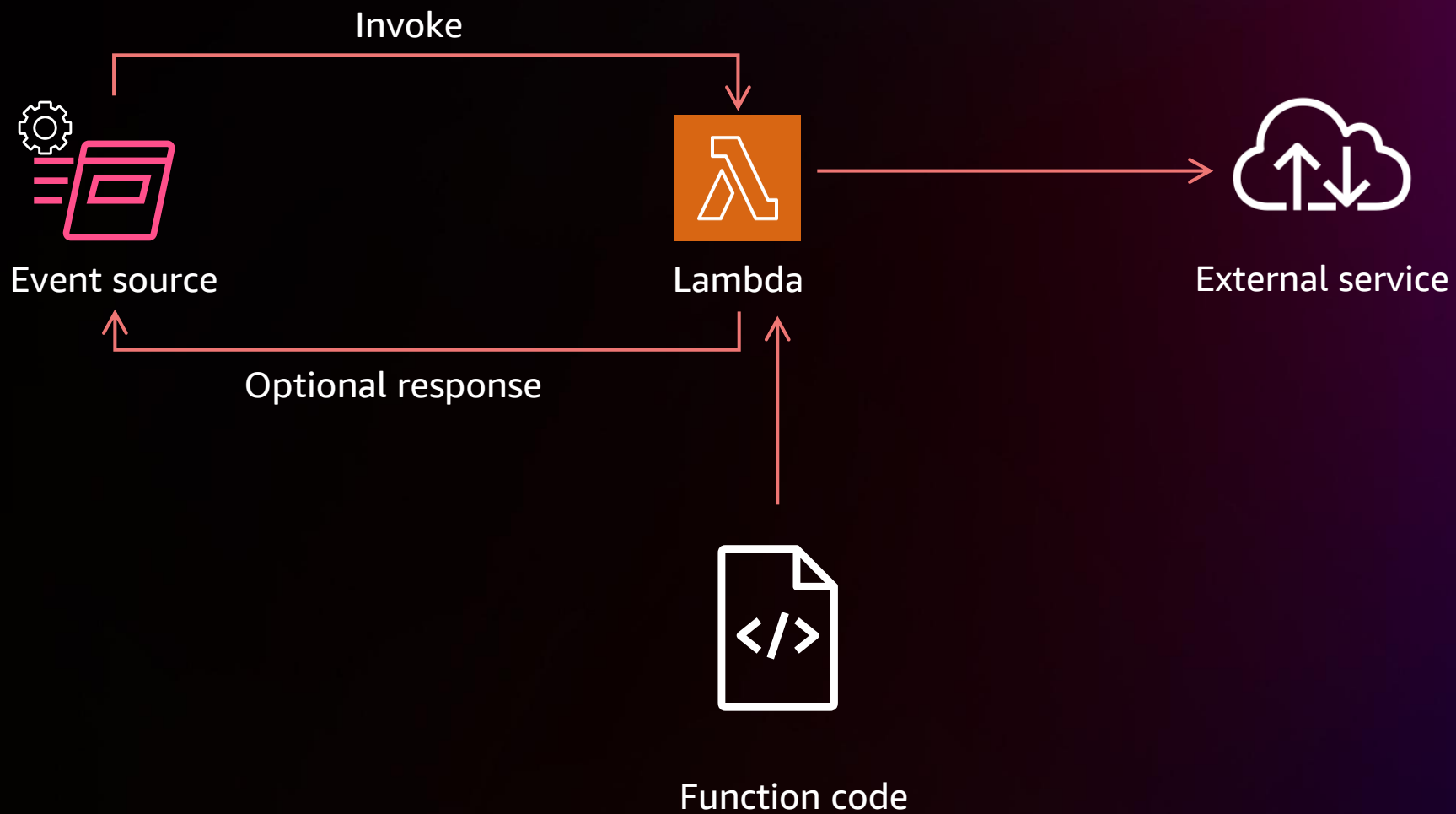


# Introduction to Lambda

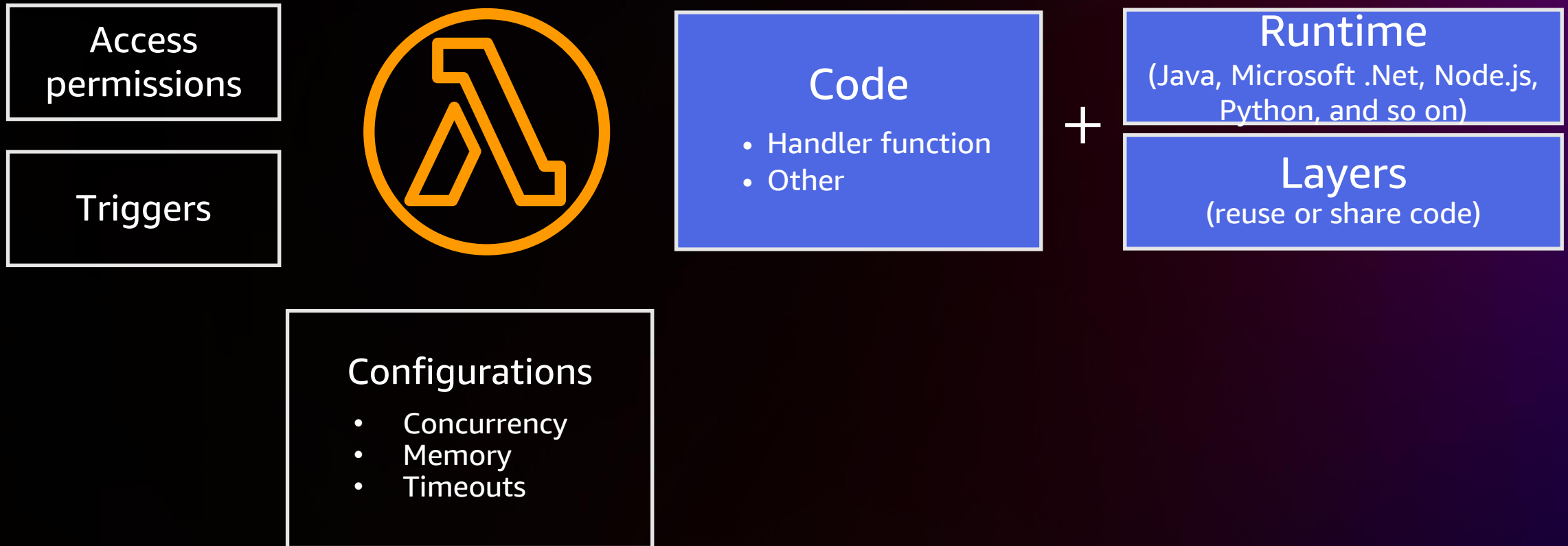




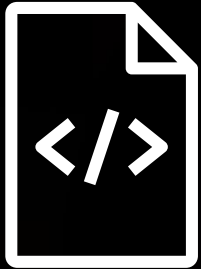
# Lambda – a compute service



# Anatomy of a Lambda function

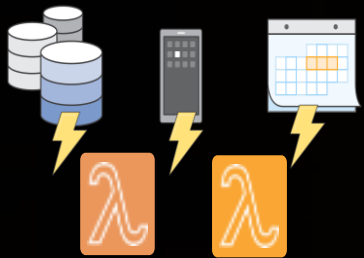


# Using Lambda



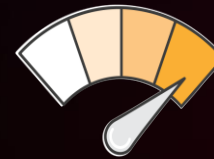
## Bring your own code

- Use Node.js, Java, Python, .NET, Go, Ruby, and others
- Create custom runtimes
- Bring your own libraries, including native libraries



## Flexible use

- Invoke in synchronous or asynchronous mode
- Integrate with other AWS services



## Simple resource model

- Select power value from 128 MB to 10 GB
- Allocate CPU and network proportionately



## Flexible authorization

- Grant access securely to resources and virtual private clouds (VPCs)
- Provide fine-grained control for invoking your functions

# Event sources that invoke Lambda

## DATA STORES



Amazon S3



Amazon  
DynamoDB

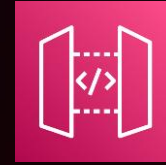


Amazon  
Kinesis



Amazon  
Cognito

## ENDPOINTS



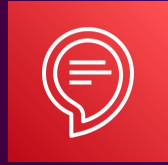
Amazon API  
Gateway



AWS IoT  
Core



AWS Step  
Functions

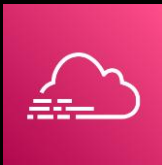


Alexa  
Smart  
Properties  
Core

## DEVELOPMENT AND MANAGEMENT TOOLS



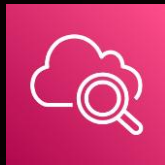
AWS  
CloudFormation



AWS  
CloudTrail



AWS  
CodeCommit

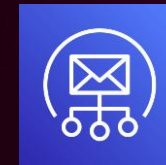


Amazon  
CloudWatch

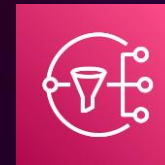
## EVENT/MESSAGE SERVICES



Amazon  
EventBridge



Amazon SES



Amazon  
SNS

... and more!

# Permissions for Lambda function

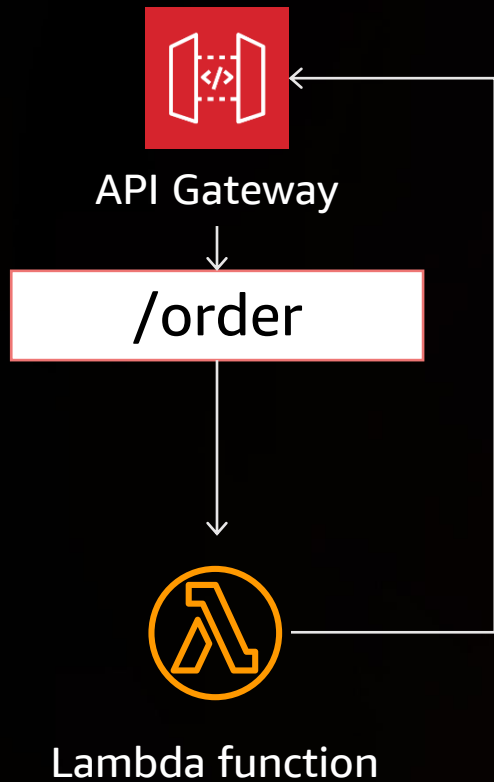
1. Push events to Lambda or invoke the Lambda function directly

2. Process the event

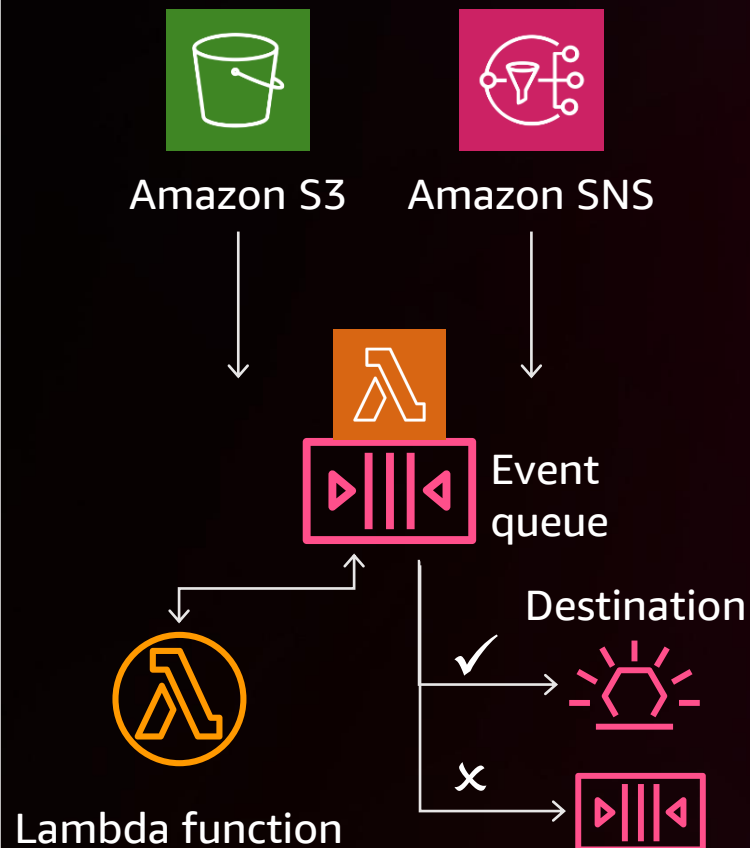


# Ways to invoke Lambda functions

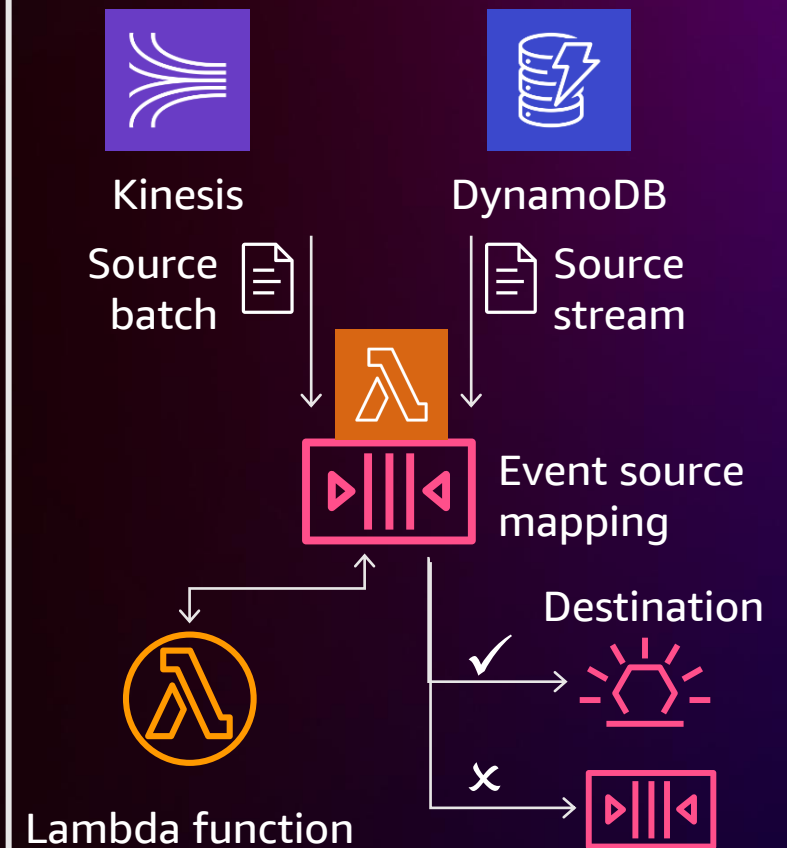
## Synchronous (direct invocation)



## Asynchronous (push)



## Poll-based (pull)

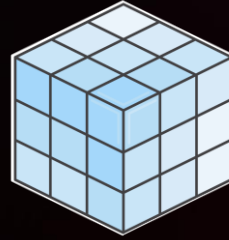


# Planning a Lambda function



## Programming model

- Use processes, threads, /tmp, sockets
- AWS SDKs



## Stateless

- Metrics for requests, errors, and throttles
- Built-in logs to Amazon CloudWatch Logs
- AWS X-Ray integration



## Monitoring and logging

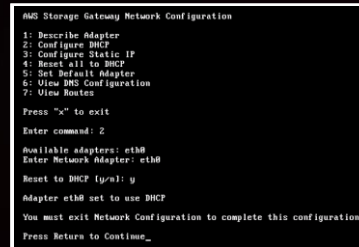
- Persist data using external storage
- No affinity or access to underlying infrastructure

# Develop and deploy

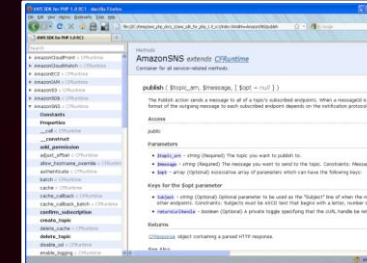
Accessible  
from



AWS Management  
Console



AWS CLI



AWS SDKs

Development  
tools





# Don't miss these learning opportunities

**AWS Jams:** Participate in teamwork-driven challenges in a fun, gamified environment or join us at the Jam Lounge in The Expo to solve daily challenges and win prizes.

**Self-paced labs:** Choose your training topic and learn as you go. Join after-hours self-paced labs on Wednesday from 6-8 pm to enjoy free pizza and beer. Or, learn on your own by accessing [reinvent.labs.aws/events.com](https://reinvent.labs.aws/events.com) remotely any time during re:Invent.

**Spotlight Labs:** Dive deeper into specific topics with AWS experts who can answer your questions.

Access the re:Invent mobile app and filter by Track: CEN: Training & Certification for specific times and locations.





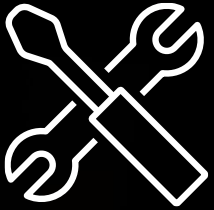
# Challenge



Start the game, create your avatar, and complete the tutorial to build your first solution and unlock your unique re:Invent challenge code.

Visit the self-paced labs room in **Venetian, Level 2, Titian 2201B** and present your code at the Cloud Quest desk to claim your prize\*!

# Expand your cloud skills with AWS Skill Builder



- 500+ Free Training Courses
- AWS Certification Practice Exams
- AWS Builder Labs
- AWS Cloud Quest
- AWS Jams

To learn more, visit the self-paced labs room at the Venetian, Level 2, Titian 2201b

# Thank you!



Please complete the session survey in the **mobile app**

